EasyA Team Maths Challenge – Question Booklet

15 January 2022, 3:00 - 3:45 P.M.

Instructions:

- This paper consists of THREE sections in order of increasing difficulty, A, B and C.
- Your team have 45 minutes to attempt as many questions as you can collaboratively, if you are in Year 13, all your attempts in session A will NOT be marked.
- To earn full marks, you must show your work on all of your answers.
- Unless otherwise stated, all answers should be given EXACTLY.
- Write your answers clearly in the spaces provided in the separated answer booklet. Remember to indicate the question that you have chosen.
- Please scan your answers as ONE PDF FILE and send them to the Team Discord Chat Channel by 4:00 p.m. Name your file according to your team's name.
- The maximum mark for this competition is **[60 Marks]**, calculated by the best questions attempted across the whole paper.
- Calculators and electronic devices are ALLOWED in this competition.

Hints for Acing the competition:

- You are allowed to look on the web if you want to recall formulas.
- There is no way you can do all the questions! Have a glance, choose the one you are most interested in, and try your best!
- You will have to decide your team's strategy!
- An incorrect answer with substantially correct calculations and explanations may still receive partial credit! Show all of your work!
- Clearly identify your answers or end of proof for each problem.

Total marks – 150.

Good luck and have fun!

Section A: GCSE Questions (5 Marks Each)

1. Question 1

Show that in any year, the 1st of April is always on the same day of the week as the 1st of July.

2. Question 2

You have 5 gold coins (Call then coin A, B, C, D, E if you want) identical in appearance. One of them is a fake coin. You are given a truthful beam balance as shown in the figure. You know the fake coin is heavier, you want to know which coin is fake.



Describe a way such that you can know the answer using the balance exactly 2 times.

3. Question 3

Find 5 numbers where, when considered together, have a mean of 18, median of 20, and mode of 22. Explain your answer.

[Hint: There is more than one choice, any choice will be acceptable.]

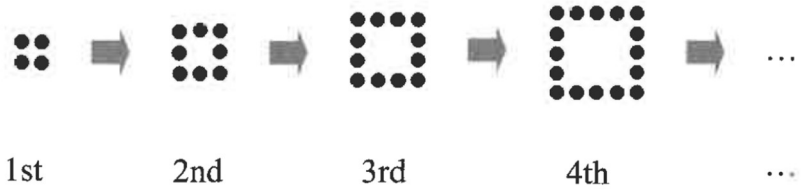
4. Question 4

From 7 A.M. to 7 P.M., the hour hand of a clock makes one full revolution. How many times do the minute hand and the hour hand overlap? Explain.

What is the time where it happens, and it is closest to 11:30 A.M.?

5. Question 5

Consider the following pattern with some dots:



Is there a pattern consisting of 82 dots? Explain.

6. Question 6

Alex is 15% taller than Ben, while Charlies is 12% shorter than Alex.

It is given that Ben is 160 cm tall.

Who is the tallest among the three people? Explain.

7. Question 7

a, b are positive numbers such that

$$\frac{6b + 5a}{9a - 5b} = 6.$$

Work out $\sqrt{a} : \sqrt{b}$.

8. Question 8

The stem-and-leaf diagram below shows the distribution of exam results of some students in a class.

Stem (tens)	Leaf (units)
6	1 1 1 3 4 6 8 9 9
7	a 7 7 8
8	1 b

It is given that the mean and the range of the above distribution is 70 and 22 respectively. Find a, b.

9. Question 9

The circle C is the circle that passes through A: $(-6,5)$ and have centre B: $(2, -1)$. Find the equation of C.

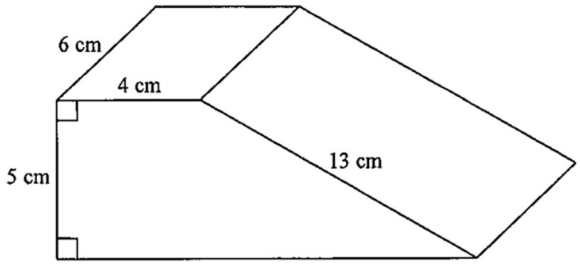
Given the point D: $(-3, 11)$ that lies outside the circle.

H is a point on the circle where it is furthest from D.

What is the relationship between points B, D, H? Find the equation of the straight line which passes through D and H. [Hint: You might find drawing a diagram here helpful.]

10. Question 10

Find the volume of the right prism shown:



[END OF SECTION A]

Section B: Easier AS Level / A Level Questions (10 Marks Each)

11. Question 11

You have a rope of length 100 m, you cut it into two parts.

You make a square and a circle with the two parts respectively.

Given the square and the circle has the same area, what is the length of the first part of the rope?

[Hint: Your answer should be in the form of

$$\frac{a}{b\sqrt{\pi}+c}$$

, where a, b, c are constants to be determined.]

12. Question 12

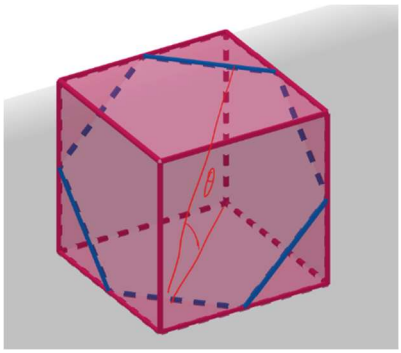
Inscribe a rectangle of base b and height h in a circle of radius 1, and inscribe an isosceles triangle in the region of the circle cut off by one base of the rectangle, with that side of the rectangle as the base of the triangle.

Draw a rough sketch of the situation described above.

For what value of h do the rectangle and triangle have the same area?

13. Question 13

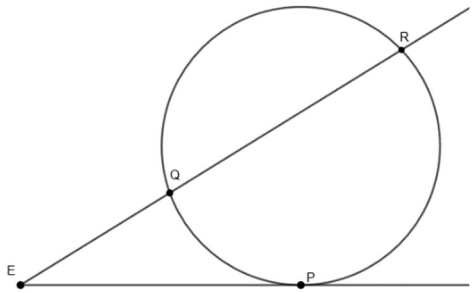
The following figure is a cube with all faces of square length 1. A hexagon (shown in blue) is formed by taking midpoints of six sides as shown. Let the angle between the base of the cube and the hexagonal surface be θ, work out $\cos\theta$.



14. Question 14

The following figure shows a circle passing through three distinct points, P, Q and R. Let E be a point lying outside the circle such that E, Q, R lies on the same straight line, show the following statement:

$QE \times RE = PE^2$ <u>if and only if</u> PE is a tangent to the circle at P.



15. Question 15

Find the complex numbers z, w that satisfies the following equation.

$$\begin{cases}(1+i)z+(1-i)w=3-i\\(1-i)z+(1+i)w=7+i\end{cases}$$

16. Question 16

20 deer live in a forest, out of which 7 are tagged. Later on, 4 random deer of the 20 are captured.

What is the probability that among those 4 captured deer, exactly 2 of the deer will be tagged?

[END OF SECTION B]

Section C: Harder A Level Questions (15 Marks Each)

17. Question 17

A sequence of polynomials $(f_0(x), f_1(x), f_2(x)\dots)$ satisfies

$$\frac{d}{dx}f_{n+1}(x) = (n+1)f_n(x+1)$$

For all positive integer n $(n \geq 1)$. Also, given

$$f_0(x) = 1, \qquad f_1(0) = f_2(0) = \cdots = 0$$

Work out the first three polynomials $f_1(x)$, $f_2(x)$ and $f_3(x)$.

Also, try to prove $f_n(1) = (1+n)^{n-1}$ for all $n \geq 1$. (*Hint:* Try to prove a statement about $f_n(x)$.)

18. Question 18

I have a calculator which only has two buttons: "+1" and "+2". The display of the calculator is at "0". Pressing "+1" makes the displayed number increase by 1, "+2" makes it increase by 2.

Here are some different ways that I can make it display the number "6".

1. Pressing "+1" 6 times.
2. Pressing "+1" 4 times, then press "+2".
3. Press "+2" First, then press "+1" 4 times.

Note that (2.) and (3.) are different ways to display the number "6".

Calculate the number of different ways I can make "1", "2", "3", "4" and "5" respectively on my calculator. You may want to list out all possibilities.

Look at the 5 numbers, do you see any patterns? Briefly Explain.

Let a_n be the number of different ways I can make the number n.

(Note: you just found a_1, a_2, a_3, a_4, a_5 previously.)

Write down a (recurrence) relation for a_n. Briefly explain why you would expect this pattern to hold for general n.

19. Question 19

This question leads you through a proof of why Euler's number e is irrational.

Given a positive integer m, define $f_m(x) = e^{-x}x^m$. We are interested about a sequence of number (integrals) defined by:

$$I_m := \int_0^1 f_m(x)\ dx$$

Again, note that $I_1, I_2, ...$ are found by evaluating the integral.

Find I_1. Now express I_{m+1} in terms of I_m using integration by parts.

Finally, try to prove

$$I_m = m!\left(1 - \frac{1}{e}\sum_{k=0}^{m}\frac{1}{k!}\right)$$

Now, you are further given $0 < I_n < \frac{1}{e}$ for all positive integer n.

Try to conclude e is irrational.

20. Question 20

Let I be the 2×2 identity matrix $\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$.

Let $M = \begin{pmatrix}p & q\\ r & -p\end{pmatrix}$, what is M^2? If $p^2 + qr = 1$, prove for all integer $n \geq 1$

$$(M + sI)^{2n} = \frac{(s+1)^{2n} - (s-1)^{2n}}{2}M + \frac{(s+1)^{2n} + (s-1)^{2n}}{2}I.$$

Let $A = \begin{pmatrix}6 & 3\\ -1 & 2\end{pmatrix}$.

Use the previous part as a hint: What is A^{800}?

Use this result to find $(A^{-1})^{800}$.

[END OF QUESTION PAPER]